

TuxShow v1.1.0: Complete User Manual

Quick Start Guide

Welcome to TuxShow! Whether you are running a school play, a live broadcast, or an experimental projection mapping project, this guide will get your first show up and running in under five minutes.

Step 1: Prepare Your Canvas

1. Look at the Header (top right) and click the green **Open Projector Screens** button.
2. This will send your blank output canvas to any connected external projectors or monitors.
3. **Tip for Classrooms:** Check the Status Bar at the very bottom of your screen. If Virtual HTTP is ON, your teacher or crew can watch the live output from any browser on the local network (e.g., <http://192.168.0.103:8554/display1>).

Step 2: Build Your Cue List

The Cue List on the left side of the screen is where your show lives. Cues are executed in order, from top to bottom.

1. Click the **+** icon in the Cue List header to create a new cue.
2. Organize your show by clicking the **Folder** icon to create groups (like a "Pre-Show Sequence" folder).
3. Drag and drop your cues to reorder them.

Step 3: Configure with the Inspector

The Inspector (bottom right) changes dynamically based on what you have selected in the Cue List. Let's set up a title slide:

1. Select a cue in your list (e.g., Cue 4).
2. In the Inspector, set the **Ref/Type** drop-down to **Text / Title**.
3. Name your cue something descriptive in the **Cue Name** field (e.g., "Welcome Title").
4. By default, the Output Routing is set to **All Displays**. Leave it here for now.
5. Watch the Stage Preview (top right) to see exactly what your audience will see in real-time.

Step 4: Run the Show!

At the bottom of the Cue List are your main playback controls.

1. Press the giant green **GO** button to fire your selected cue.
2. Notice the Active Playhead indicator in the top header. TuxShow allows multiple cues to run simultaneously (for example, firing a Background Video and Ambient Music at the same time).

3. Need to halt everything immediately? Hit the red **PANIC (!)** button to gracefully fade out and stop all active cues.

Step 5: Pack It Up

When class or rehearsals are over, you need to take your show with you.

1. Go to the Header and click **Pack**.
2. TuxShow will bundle your `.TSW` workspace file and all linked media into a portable folder so you can take it anywhere.

Chapter 1: Welcome to TuxShow

What is TuxShow?

Welcome to TuxShow, the accessible, powerful live event control and projection mapping application. Built specifically with students, teachers, and school drama departments in mind, TuxShow bridges the gap between basic slideshow software and highly complex, expensive industry-standard tools (like QLab or Resolume).

Whether you are running sound cues for a middle school play, projecting dynamic video backdrops for a high school musical, or teaching students the fundamentals of projection mapping, TuxShow provides a professional feature set in a cost-effective, easy-to-learn package.

System Requirements & Installation

Because TuxShow relies on live video rendering and WebGL for effects, we recommend the following minimum specifications:

- **OS:** Ubuntu / Debian, or Raspberry Pi OS (optimized for Raspberry Pi 4 and 5).
- **Processor:** Intel Core i5 / AMD Ryzen 5, or Raspberry Pi 4/5 ARM processors.
- **RAM:** 8GB minimum (16GB recommended for heavy video playback).
- **Graphics:** A dedicated GPU is recommended for multiple 1080p outputs or complex projection mapping.
- **Installation:** Simply download the appropriate installation package (e.g., `.deb`) for your operating system, run the setup, and launch the application. No complex licensing dongles are required.

License

TuxShow is proud to be free and open-source software. This project is licensed under the GNU General Public License v3.0 (GPLv3). This ensures that TuxShow remains freely available for educational institutions, theaters, and developers to use, modify, and distribute, fostering a collaborative community around live event technology.

Understanding the Unified Timing Architecture

One of the most confusing aspects of professional show control software is managing complex timelines. TuxShow solves this with a **Unified Timing Architecture**.

Instead of dragging blocks around a multi-track timeline, every single cue in TuxShow—whether it's a video, an audio track, a text title, or a blackout—shares the exact same timing logic, configured directly in the Inspector:

- **Fade In:** How long (in seconds) the cue takes to appear or reach full volume.
- **Duration:** How long the cue runs. (Leave blank for infinite loops).
- **Fade Out:** How long the cue takes to disappear or fade to silence.
- **Follow Action: * Wait for GO:** The show pauses after this cue and waits for the operator to press the GO button.
 - **Auto-Follow:** As soon as this cue finishes its duration, the next cue in the list triggers automatically.

Chapter 2: The User Interface (A Guided Tour)

The TuxShow interface is designed for high-stress live environments. It is dark-themed to reduce eye strain in dimly lit tech booths and is divided into five main functional areas.

1. The Header (Top)

The top bar of the application is your global command center.

- **File Controls:** Standard buttons for New, Load, Save, and Pack.
- **Playhead Status:** Clearly displays your Selected Cue (the cue you are currently editing) and the Active Playhead (a list of all cues currently running live on stage).
- **Projection & Mapping:** Contains tools to tweak your outputs, including the Map Surface toggle.
- **Open Projector Screens (Green Button):** Clicking this instantly sends your output canvas to any connected external monitors or projectors.

2. The Cue List (Left Column)

This is the script for your show. Cues execute in order from top to bottom.

- **Organization:** You can drag and drop cues to reorder them, use the Search bar to find specific items, or click the Folder (+) icon to group cues together (e.g., creating a "Pre-Show Sequence" folder).
- **Visual Indicators:** Each cue displays a specific icon (Video, Audio, Text, etc.), its name, and running timers. Active cues are highlighted with a green outline and a "Play" triangle.
- **Playback Controls (Bottom Left):**
 - **GO (Large Green Button):** Fires the currently selected cue and advances the playhead.
 - **PAUSE (Yellow Button):** Pauses all currently running media.

- **PANIC (Red ! Button):** The emergency stop. Instantly fades out and stops all active cues.

3. Stage Preview (Top Right)

This live 2D canvas shows you exactly what is being sent to the projectors. If you have a video cue running (like a nature background) and a text cue layered on top of it, you will see the composited result here in real-time. If you click the Map Surface button in the header, this preview window allows you to drag the corner pins to warp the main output grid (Affine Transform).

4. The Inspector (Bottom Right)

The Inspector is context-sensitive; its contents change completely depending on what cue you have selected in the Cue List.

- **Cue Details:** Set the Cue Number, Cue Type (e.g., Text/Title), and Cue Name. You can also assign the cue to a Parent Group.
- **Output Routing:** Choose whether this specific cue goes to All Displays or a specific connected projector.
- **Geometry, Crop & Outline:** For visual cues, this is where you manipulate the image. You can adjust Scale X/Y (with a "Keep Aspect Ratio" checkbox), move the media using Pos X/Y percentages, or crop the bounds.

5. The Status Bar (Bottom)

This crucial strip runs along the very bottom edge of the window, providing real-time network and hardware diagnostics:

- **Host IP:** Displays the current IP address of your computer (e.g., 192.168.0.103).
- **Virtual HTTP:** Shows if your local network broadcast server is running and provides the exact URL (e.g., <http://192.168.0.103:8554/display1>) that students can use to view the show on their own devices.
- **OSC/MSR RX:** Shows the active port for incoming Open Sound Control messages (e.g., PORT 53000) and the status of MIDI Show Control inputs.

Chapter 3: File Management & Portability

Understanding .TSW Files

At its core, a TuxShow workspace is saved as a .TSW (TuxShow Workspace) file. Under the hood, this file is a structured JSON payload. It contains:

- The order, hierarchy, and properties of all your cues.
- Unified timing parameters (Fade In, Duration, Fade Out, Follow Actions).
- Coordinate and configuration data for projection mapping and warps.

- File paths pointing to your media assets.

Because it is a text-based JSON file, a `.TSW` file itself is incredibly small and easily shared. However, it does not inherently contain your actual video, image, or audio files—it only contains links to where they are stored on your system.

The Absolute Path Limitation

By default, when you drag and drop media into TuxShow, the application records their absolute file paths (e.g., `/home/pi/Documents/MyShow/media/background.mp4` on Raspberry Pi, or `/home/user/Desktop/video.mp4` on Ubuntu).

While this works perfectly while you are programming on a single computer, it presents a problem if you move your `.TSW` file to a different machine. If the second machine does not have the exact same files stored in the exact same directories, the media links will break, and TuxShow will show empty/offline placeholders.

The Solution: The "Pack Workspace" Feature

To make your show fully portable, TuxShow includes a powerful bundling engine. By packing your workspace, you convert all absolute paths to relative paths and aggregate all resources.

How to Pack a Show:

1. Open your running workspace in TuxShow.
2. In the top Header, click the **Pack** button.
3. Choose a destination folder (such as an external USB thumb drive).
4. TuxShow will automatically:
 - Create a new folder at your destination.
 - Generate a newly structured workspace file named `packed_show.TSW`.
 - Create a `./media/` subfolder immediately adjacent to this file.
 - Copy every active video, image, and audio asset used in your show into that `./media/` folder.
 - Re-write all file references inside `packed_show.TSW` to use relative paths (pointing to `./media/`).

This bundled folder is completely self-contained. You can plug your USB drive into any other Ubuntu, Debian, or Raspberry Pi running TuxShow, open `packed_show.TSW`, and run your show immediately without any broken links.

Chapter 4: Building Your Show: Cue Types

The Inspector allows you to change any blank or existing cue into one of several dynamic types. This chapter details each cue type, its primary purpose, and its configuration parameters.

1. Media Cues

Video & Audio

- **Purpose:** High-performance, synchronized playback of visual and auditory files.
- **Key Settings:**
 - **Volume Slider:** Adjusts output volume from 0% to 100%.
 - **Loop Toggle:** When enabled, the file repeats indefinitely until stopped by another cue or a PANIC command.
 - **Continuous Play:** Keeps the audio or video track running in the background even when the visual element is hidden or the slide changes.

Images

- **Purpose:** Displays static scenery, lower thirds, or graphic announcements.
- **Key Settings:** Supports basic scale, position, opacity, and timing. Loop functions are unnecessary for static images as they remain on screen for their designated Duration.

2. Live Camera Cues

- **Purpose:** Pulls live video feeds directly onto the stage canvas. This is perfect for stage monitoring, live stream overlays, or digital scenic integrations.
- **Supported Sources:**
 - **Local USB Webcams:** Select standard video inputs connected directly to your computer or Raspberry Pi.
 - **Network Streams:** Enter network URLs to pull remote feeds using common industrial protocols like RTSP (rtsp://), HTTP MJPEG feeds, or Open Media Transmitter (OMT) protocols.

3. Text/Titles Cues

- **Purpose:** A generative text engine that renders custom typography on the fly without needing external graphic editing software.
- **Key Settings:**
 - **Content Box:** Type the text exactly as you want it to appear (supports multi-line text).
 - **Font & Styling:** Choose standard system fonts, adjust sizes, weights, alignments, and pick custom colors.
 - **X/Y Position:** Place the text anchor precisely on the stage layout using percentages (0% to 100%).
 - **Drop Shadow:** Customize the shadow offset, blur, and color to ensure your titles remain readable against changing background videos.

4. Logic & Control Cues

Group (Folders)

- **Purpose:** Organize your show into structural blocks (e.g., "Act 1, Scene 1", "Intermission Loops").
- **Key Features:** Beyond organization, groups can be set to **Fire All**. When a Group cue is triggered with "Fire All" enabled, it will execute all child cues contained inside the folder simultaneously.

GoTo & Counter

- **Purpose:** Create loops, conditional sequences, or non-linear jumps in your show.
- **Key Settings:**
 - **Target Cue:** Specify the cue number you want the playhead to jump to.
 - **Counter Limit:** Set how many times this jump is allowed to execute. Once the limit is reached, TuxShow will ignore the jump and move down to the next sequential cue. This is highly useful for running a background loop exactly three times before automatically proceeding to the next scene.

Time

- **Purpose:** Automate your show based on real-world scheduling.
- **Key Settings:** Program a precise date and clock time (down to the second). The application will wait and trigger this cue autonomously. This is widely used for museum installations or automated school bell systems.

Stop & Pause

- **Purpose:** Programmatic management of playing media.
- **Key Settings:**
 - **Target Cue Number:** Choose a specific cue to target.
 - **Stop Action:** Instantly halts and unloads the target media.
 - **Pause Action:** Freezes the target media in place, allowing a future trigger to resume it exactly where it was halted.

Blackout

- **Purpose:** Fades the entire stage output canvas to pure black.
- **Key Settings:** Governed by your Fade In time. If you configure a 3-second Fade In, the entire stage will elegantly fade to black over 3 seconds.

Transition

- **Purpose:** Overlay dynamic mathematical wipes and graphic transitions on top of active video outputs.
- **Supported Styles:** Built-in WebGL-driven transitions including Iris-in, Wipe-Up, and Ripple effects. Perfect for stylized scene changes.

Chapter 5: Advanced Visuals & Projection Mapping (New in v1.1.0)

When a visual cue (such as a Video, Image, Camera, or Text cue) is selected in the Cue List, the Inspector unlocks a suite of powerful geometric rendering tools. These features allow you to warp, mask, and color-correct individual pieces of media to perfectly align with physical stage sets, columns, or unconventional projection surfaces.

1. Basic Geometry, Crop & Outline

Before jumping into advanced warping, use the core geometric modifiers to position your media:

- **Scale X/Y & Keep Aspect Ratio:** Adjust the size of your visual cue. Keep the Keep Aspect Ratio checkbox enabled to prevent stretching or squishing actors and graphics.
- **Position X/Y:** Moves the center anchor of the media around the screen. This uses a percentage grid from 0% to 100%. For example, setting both values to 50 perfectly centers the media on the stage.
- **Crop Bounds (Top/Bottom/Left/Right):** Trim unwanted edges of your media directly in the software. This is highly useful for removing black bars from videos or isolating a specific part of a live camera feed.
- **Outline:** Add a vector stroke border around your media. You can customize both the color and the pixel thickness of this outline, making it easy to create highlighted visual frames for live feeds or text cards.

2. Perspective Warp (Per-Cue Affine Transform)

Unlike the global master grid alignment, TuxShow allows you to warp individual cues independently. This means you can project a background loop across the entire stage while simultaneously warping a specific video of a "living portrait" directly onto a physical picture frame hanging on your set.

How to Warp a Cue:

1. Select your visual cue and locate the Perspective Warp section in the Inspector.
2. Toggle **Enable Warp** ON.
3. In the Stage Preview window, four blue corner pins will appear over your selected media.
4. Drag these four corners independently. TuxShow utilizes WebGL hardware acceleration to perform a live Affine Transform, mathematically stretching and squeezing the texture to match whatever quad shape you draw.

5. **Raspberry Pi Tip:** Because these warps run directly on the Raspberry Pi 4/5 GPU, they are highly performant. However, to maintain 60 FPS, avoid running more than 3 warped 1080p video cues simultaneously on Pi hardware.

3. Masking

If your projection surface isn't a perfect rectangle (for example, an arched doorway or a circular set piece), you can use the Masking tool to cut away parts of your media.

- Under the Masking menu, click **Draw Custom Mask**.
- In the edit popup, click to create points and draw a custom multi-point polygon shape directly over a preview of your media.
- Once saved, TuxShow uses this polygon to define transparency. Everything outside your drawn shape is instantly cut out, allowing your background layers to show through.

4. Chroma Key (Green Screen Removal)

TuxShow includes a real-time WebGL chroma key shader, allowing you to use live actors on green screens or play special effect video assets with green backgrounds without needing pre-rendered transparency.

- **Similarity Slider:** Adjusts how close a color must be to the target green to be removed.
- **Smoothness Slider:** Softens the edges of the key to prevent harsh pixelated lines around hair or clothing.

Chapter 6: Output Routing & Networking

Once your show is programmed, you must configure how the video signals leave your host computer and how external controls interface with TuxShow.

1. Hardware Routing & Dual-Monitor Setup

TuxShow is optimized for environments with multiple displays. When configuring your physical space, we recommend:

- **Main Monitor:** Runs the TuxShow control application (desktop interface).
- **Projector / Secondary Display:** Receives the clean theatrical output.

Raspberry Pi Configuration: The Raspberry Pi 4 and 5 both feature dual micro-HDMI ports (HDMI-0 and HDMI-1). Connect your control monitor to HDMI-0 and your theatrical projector to HDMI-1.

Inside the Inspector, you can set the **Output Routing** drop-down menu for each cue to **All Displays**, or restrict it to a specific external screen (e.g., Display 2 or Display 3). This allows a single operator to send scenery to a stage projector while sending confidence monitor feeds or backstage instructions to secondary displays.

2. Virtual HTTP Display (Network Broadcasting)

One of the most powerful educational features in TuxShow is the built-in Node.js network broadcaster. This allows you to stream your live stage canvas output over your local school network.

- **How it works:** When TuxShow starts, it registers a web stream on your local IP address (shown in the Status Bar, e.g., <http://192.168.0.103:8554/display1>).
- **In the Classroom:** Teachers, designers, or student crew members can open VLC Media Player, OBS, or a standard web browser on their own laptops or mobile devices, enter this URL, and watch the live stage projection in real-time. This is perfect for remote design critiques, virtual stage-managing, or overflow viewing areas.

3. Show Control Integration (OSC & MSC)

TuxShow is built to play nicely with professional theatrical consoles and network infrastructure.

Open Sound Control (OSC)

- **Receiving Commands:** TuxShow listens constantly for incoming OSC commands over network IP. By default, it listens on Port 53000. Other computers, mobile tablets running custom layout apps (like TouchOSC), or professional sound consoles can send commands to:
 - [/tuxshow/go](#) - Triggers the main GO button.
 - [/tuxshow/panic](#) - Instantly stops all playback.
 - [/tuxshow/cue/{number}/fire](#) - Fires a specific cue number instantly.
 - *(Note: A complete, exhaustive list of all compatible OSC commands and parameters can be found in Chapter 7: Appendices).*
- **Sending Commands:** You can create dedicated OSC Cues in your cue list. When fired, these cues send outbound network packets to trigger changes in external sound consoles (like Behringer X32/M32) or other video servers.

MIDI Show Control (MSC)

For lighting-to-video synchronization, TuxShow supports MIDI Show Control.

- Connecting a compatible USB-MIDI interface to your Ubuntu machine or Raspberry Pi allows a lighting console (like an ETC EOS or GrandMA) to automatically trigger your video cues.
- When the lighting board runs "Cue 5" on the light board, it sends an MSC "GO 5" command over MIDI. TuxShow reads this command and automatically fires "Cue 5" in your cue list, keeping sound, lights, and projections perfectly in sync with a single button press.

Chapter 7: Appendices

Appendix A: Exhaustive OSC Command Reference

TuxShow features a robust, low-latency Open Sound Control (OSC) engine. By default, the application listens for incoming UDP packets on the incoming port configured in your I/O Settings (default: Port 53000). To control TuxShow from third-party software (such as QLab, TouchOSC, Max/MSP, or hardware controllers), use the following syntax and address paths.

1. Global Transport & Show Control

These control the primary show state, identical to the hardware buttons at the bottom of the cue list.

OSC Address Path	Argument Type	Description
/tuxshow/go	None	Triggers the currently selected cue(s) (equivalent to pressing the physical spacebar or green GO button).
/tuxshow/stop	None	Hard-stops all currently active cues immediately, ignoring programmed fade-out times.
/tuxshow/pause	None	Pauses the global auto-follow timeline timers and freezes all running video/audio playback.
/tuxshow/resume	None	Unpauses the global timeline and resumes playback of all playing media.
/tuxshow/panic	None	Triggers a soft stop on all active cues, allowing them to gracefully fade out using their programmed fadeOut Time parameters.

2. Selection & Playhead Controls

These allow external control pads (such as TouchOSC or a Stream Deck) to navigate the cue list.

OSC Address Path	Argument Type	Description
/tuxshow/select/next	None	Shifts the active selection highlight down to the next visible cue.
/tuxshow/select/prev	None	Shifts the active selection highlight up to the previous visible cue.
/tuxshow/select/cue	String or Float	Jumps the selection highlight directly to the specified cue number (e.g., "1.5" or 2) and scrolls it into view.

3. Master Controls

OSC Address Path	Argument Type	Description
/tuxshow/master/volume	Float (0.0-1.0)	Dynamically scales the overall audio volume across the entire engine.

4. Targeted Cue Actions (Dynamic Wildcard)

These allow you to manipulate specific individual cues dynamically by their user-defined cue numbers, bypassing the active selection highlight. The format is

/tuxshow/cue/[cue_number]/[action]:

OSC Address Path	Argument Type	Description
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/tuxshow/cue/[cue_number]/start	None	Remotely fires the targeted cue. If the cue is inside a folder or has auto-follows, it will process down the cue stack.
/tuxshow/cue/[cue_number]/stop	None	Soft-stops only the targeted cue, respecting its individual fadeOutTime.
/tuxshow/cue/[cue_number]/pause	None	Pauses the underlying HTML media player element of the targeted cue.
/tuxshow/cue/[cue_number]/resume	None	Resumes the underlying HTML media player element of the targeted cue.
/tuxshow/cue/[cue_number]/volume	Float (0.0-1.0)	Updates the local volume parameter of the targeted cue in real time.
/tuxshow/cue/[cue_number]/opacity	Float (0.0-1.0)	Dynamically adjusts the visual layer opacity of the targeted cue on the master stage canvas.

5. Outgoing Commands (Triggered via OSC Cues)

When you add an OSC Cue to your TuxShow cue list, the Inspector allows you to define:

- **Target IP:** The network address of the destination device (e.g., 192.168.0.105).

- **Target Port:** The UDP port the destination device is listening on (e.g., Port 8000 for a Behringer digital mixer or 53000 for QLab).
- **Address Path:** Custom string (e.g., /ch/01/mix/fader).
- **Arguments:** Supports sending single or multiple dynamic arguments consisting of Strings, Floats, or Integers.

Appendix B: Keyboard Shortcuts

IMPORTANT NOTE FOR OPERATORS: Keyboard shortcuts are not implemented in the current v1.1.0 release. The mapping table below outlines the planned shortcut specifications designed for a future version of TuxShow to support rapid, mouse-free theatrical operation.

Category	Planned Shortcut (Linux / Pi)	Intended Action
Playback	Spacebar	GO (Triggers the selected cue)
Playback	Escape	PANIC (Fades out and stops all active cues instantly)
Playback	P	Toggle global Play/Pause
Navigation	Up Arrow	Select the previous cue in the list
Navigation	Down Arrow	Select the next cue in the list
Navigation	Ctrl + F	Focus on the Cue Search bar
System	Ctrl + N	Create a New Workspace

System	Ctrl + O	Open/Load a .TSW file
System	Ctrl + S	Save current Workspace
System	Ctrl + Shift + P	Open the "Pack Workspace" dialog
Projection	M	Toggle Map Surface mode (edit master warp grid)
Projection	F11	Toggle fullscreen window mode

Appendix C: Theatrical & Technical Glossary

- **Affine Transform:** A geometric rendering method used in TuxShow's perspective warp. It mathematically stretches, rotates, and translates a 2D visual asset to fit custom 4-point coordinates without bending straight lines.
- **Chroma Key:** A real-time green screen removal tool. It filters out green backgrounds from cameras or videos using WebGL shaders, making them transparent so you can layer backgrounds underneath.
- **Cue:** An individual action or event (such as playing a sound, starting a video, displaying text, or sending a network command) triggered by the operator.
- **Follow Action:** A setting that tells TuxShow what to do after a cue completes. It can wait for a manual operator trigger ("Wait for GO") or run the next cue instantly ("Auto-Follow").
- **MIDI Show Control (MSC):** An industry-standard extension of MIDI protocol that allows lighting consoles, sound software, and media servers to communicate cue triggers over a physical interface.
- **Open Sound Control (OSC):** A high-performance, network-based message protocol used to connect show-control software and hardware over standard Wi-Fi or Ethernet.
- **Playhead:** The visual marker indicating which cue is currently selected or running.
- **Projection Mapping:** The art and technique of warping projections to fit perfectly onto physical, non-flat surfaces like theatrical sets, pillars, or structures.
- **TSW File:** The proprietary file extension (.TSW) for TuxShow Workspaces. This is a lightweight JSON file containing all structural settings, mapping coordinates, and absolute file paths.

Appendix D: Technical Troubleshooting

Running live video effects on student hardware requires structured troubleshooting. Below are solutions to the most common issues encountered on Ubuntu, Debian, and Raspberry Pi 4/5.

1. The Projector Display is Blank or Black

- **Is the output window open?** Ensure you clicked the bright green Open Projector Screens button in the header.
- **Check OS Screen Arrangement:** On Ubuntu or Raspberry Pi OS, open your system's Display Settings. Verify that your desktop is set to Join Displays (Extend Desktop) rather than "Mirror/Clone."
- **HDMI Port Order (Raspberry Pi Only):** The Raspberry Pi 4/5 features two micro-HDMI ports. HDMI-0 (the port closest to the USB-C power input) must go to your primary control monitor. HDMI-1 should be connected to your projection system.

2. Video Playback is Stuttering or Dropping Frames

- **GPU Memory (Raspberry Pi 4):** By default, Raspberry Pi 4 may limit video memory allocation. Open your terminal, run `sudo raspi-config`, navigate to Performance Options -> GPU Memory, and set it to at least 128MB or 256MB before rebooting. (Note: The Raspberry Pi 5 handles this dynamically, so this step is unnecessary on Pi 5).
- **Optimize Media Assets:** High-bitrate 4K videos will overwhelm Raspberry Pi systems. For buttery-smooth playback, encode all video files as 1080p H.264 (MP4) with a bitrate under 15 Mbps.
- **Limit Perspective Warps:** Warping uses heavy real-time WebGL calculations. On standard Raspberry Pi hardware, try to limit active perspective warps to no more than 3 simultaneous video layers.

3. OSC or MSC Commands are Not Triggering Cues

- **Firewall Restrictions:** Debian and Ubuntu systems often have active firewall utilities (`ufw`). Open your terminal and run `sudo ufw allow 53000/udp` to allow incoming OSC traffic on TuxShow's default port.
- **IP Conflicts:** Double-check the Status Bar at the bottom of the screen. Ensure the sending device (e.g., lighting console or tablet) is sending commands to the exact Host IP shown in TuxShow. Both devices must be connected to the exact same router/subnet.
- **MIDI Device Permissions (Linux):** If using a USB-MIDI interface for MSC, your Linux user account might lack permission to read direct MIDI devices. Open your terminal and add your user to the audio group:
 - `sudo usermod -a -G audio $USER`
 - (Log out and log back in for changes to take effect).

4. "Media Offline" / Red Warning Symbols on Cues

- **Broken File Paths:** This occurs because the `.TSW` file was moved to a different computer, but the linked media assets were left behind on the original machine.

- **The Fix:** Go back to the original computer, open the workspace, and click the Pack button in the header. Move the newly generated folder containing `packed_show.TSW` and the `./media/` directory to your thumb drive. This ensures all file paths remain relative and functional.